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VETFUSION

SMART GLOVE FOR REAL-TIME WOUND AND INFECTION DETECTION

PROJECY GUIDE : Mrs. P.SARANYA

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ABSTRACT

This project presents a smart wearable Glove system for real-time wound and infection detection in dogs. Temperature, humidity, and color sensor data are acquired using ESP32 DevKit V1, while wound images are captured using ESP32-CAM. The RGB LED provides instant indication based on sensor output during measurement. Sensor data is processed using noise removal and threshold techniques to improve accuracy. Both sensor data and image data are integrated with a pre-trained machine learning model for classification of conditions such as healthy, inflammatory infection, ringworm, and demodicosis. The processed data is sent to a cloud platform and displayed in a mobile application for real-time monitoring. The system is portable, low-cost, supports early detection of infections.

LITERATURE SURVEY

S. No	Title	Journal / Year	DOI / ISSN	Influence	Drawback
1	Convolutional Neural Networks for Wound Detection: The Role of AI in Wound Care	Journal of Wound Care / 2019	10.12968/jowc.2019.28.Sup10.S13	Guides use of CNN for wound detection.	Human-focused; paywalled
2	Dog Skin Disease Recognition Using Image Segmentation and GPU Enhanced Convolutional Neural Network	IEEE / 2022	10.1109/ACCESS.2022.3194529	Provides insights into segmentation and classification of canine skin diseases.	Focused on specific skin diseases; may not cover all canine conditions.
3	AI-driven veterinary diagnostics	IEEE J. Biomed. Health Inform. / 2024	10.1109/JBHI.2024.3361234	Discusses AI in veterinary diagnostics and mobile app integration.	Paywall; wound focus unclear
4	Artificial Intelligence in Veterinary Care: A Review of Applications for Animal Health	Egyptian Journal of Veterinary Sciences / 2024	10.21608/ejvs.2024.260989.1769	Reviews AI applications in veterinary diagnostics and treatment.	General overview; lacks specific case studies
5	Automatic wound detection and size estimation using deep learning algorithms	PLOS Computational Biology / 2022	10.1371/journal.pcbi.1009852	Provides an end-to-end deep learning pipeline for wound analysis.	Focused on mouse models; may not generalize to other species

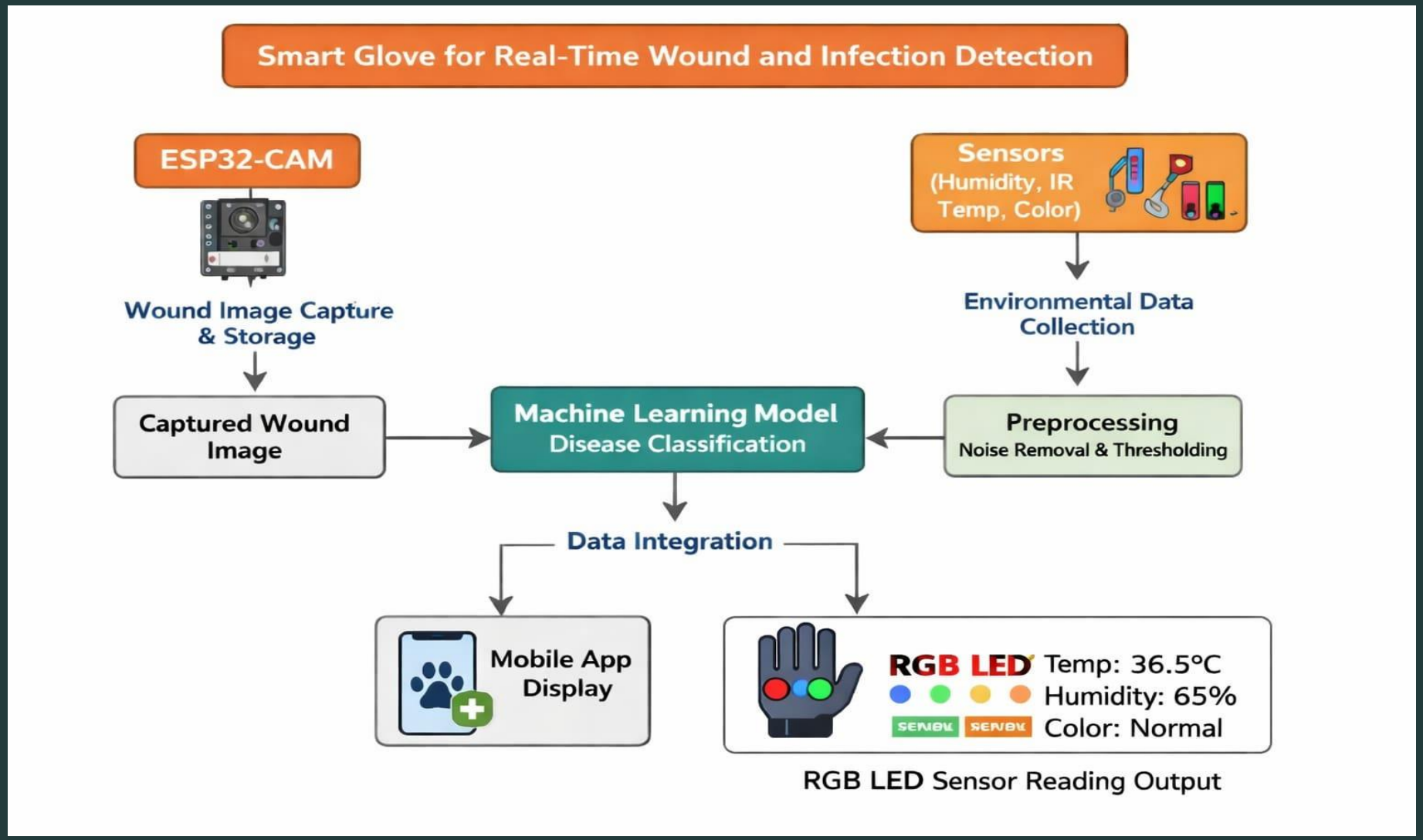
PROBLEM STATEMENT

- Increasing wound infections in dogs require early detection and continuous monitoring
- Existing methods rely on manual inspection, which is time-consuming and less accurate
- Lack of low-cost, portable, and user-friendly systems for easy use in real-world conditions
- Available devices are bulky and not suitable for wearable or field applications
- Limited integration of hardware and software with cloud and mobile connectivity for real-time monitoring

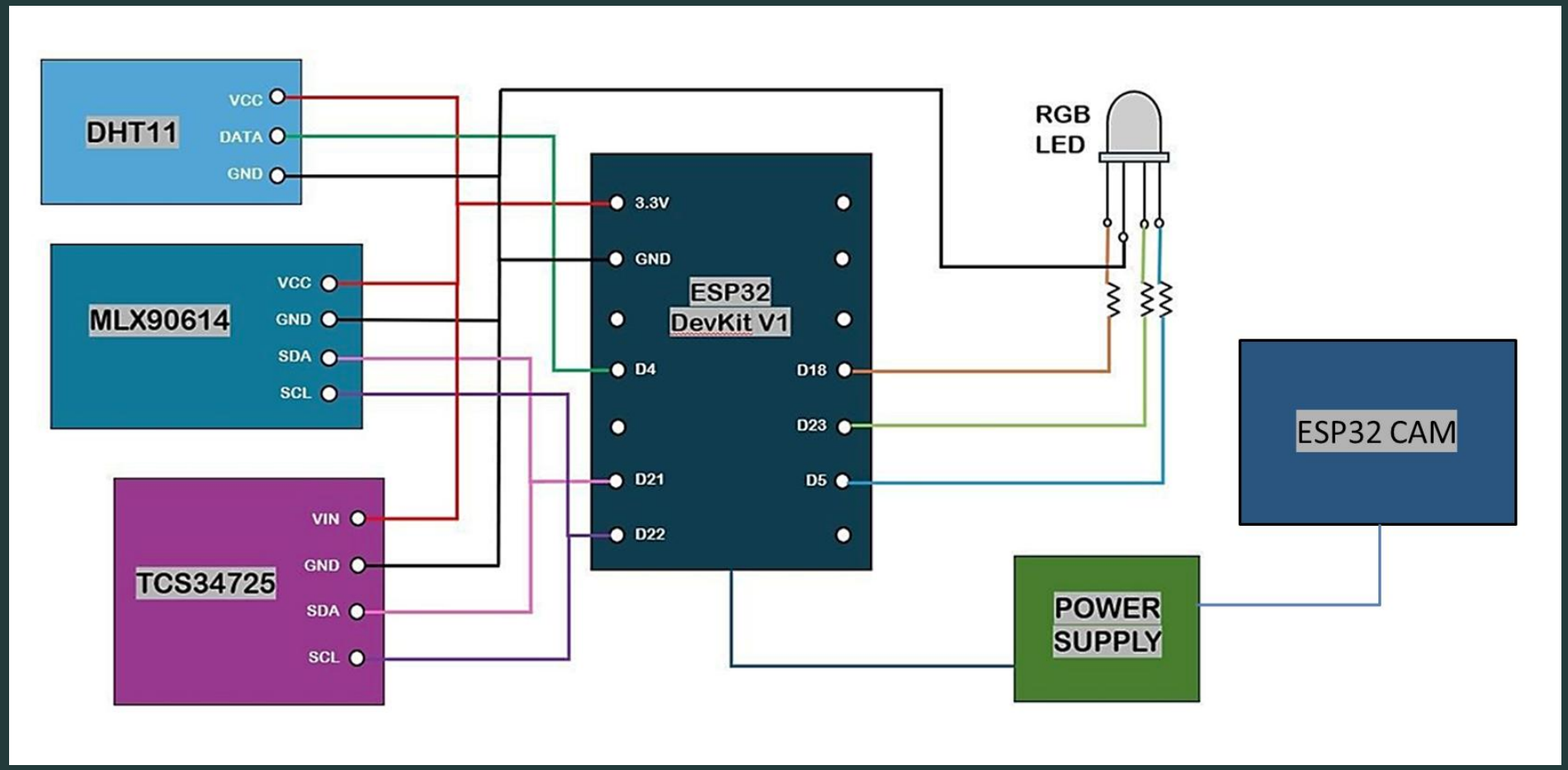
💡 PROPOSED SOLUTION

- Develop a smart wearable Glove system using ESP32 DevKit V1 to acquire temperature, humidity, and color sensor data from the dog
- Capture wound images using ESP32-CAM for visual analysis
- Provide instant RGB LED indication in real-time while sensing based on sensor output
- Apply noise removal and threshold techniques on sensor data to improve accuracy
- Integrate sensor data and ESP32-CAM images with a pre-trained machine learning model and send results to a cloud platform for display in a mobile application

PROPOSED SYSTEM ARCHITECTURE



CIRCUIT DIAGRAM



HARDWARE WORKING

- An FTDI Programmer is used to upload code to the ESP32- CAM
- ESP32-CAM captures dog wound image
- Sensors (MLX90614, DHT11, TCS34725) are connected to the ESP32 Devkit V1
- Temperature, humidity, and color data are collected from the dog
- ESP32 Devkit V1 processes sensor values using threshold logic
- RGB LED indicates status (Green: Normal, Red: Infection, Blue : Mild)
- Sensor data and images are sent to the next software Machine learning stage

SOFTWARE WORKING

ESP32 Programming & Data Collection: (ARDUINO IDE)
ESP32-CAM and ESP32 DevKit V1 are programmed using Arduino IDE to capture images and sensor data (temperature, humidity, RGB).

Image Processing: (VISUAL STUDIO CODE)
Captured images are processed to improve quality using noise removal, resizing, and thresholding.

Machine Learning Classification : (GOOGLE COLAB)
Google Colab is used to train and test the model to classify Healthy, Demodicosis, Ringworm, and Inflammatory Infection.

Cloud Integration : (THINGSPEAK)
Sensor data, images, and prediction results are uploaded and stored in real time.

Mobile Application Display : (MIT APP INVENTOR)
MIT App Inventor is used to develop a mobile app for real-time visualization of results and sensor data.

METHODOLOGY

DATA ACQUISITION → THRESHOLD BASED INDICATION AND PREPROCESSING → DATA STORAGE → MACHINE LEARNING INTEGRATION → CLOUD ,MOBILE APP AND GLOVE INTEGRATION

1. **DATA ACQUISITION (ESP32) :** ESP32-CAM captures dog images and ESP32 DevKit collects sensor data such as temperature, humidity, and color sensor values in real time.

2. **THRESHOLD BASED INDICATION AND PREPROCESSING :** Sensor readings are analyzed using threshold values to control RGB LED indication, and noise removal techniques are applied for data cleaning.

3. DATA STORAGE : Captured images are stored in Google Drive, and sensor readings are saved as CSV files for further processing and analysis.

4. MACHINE LEARNING INTEGRATION : Image data and sensor CSV data are integrated and fed into a pretrained model to classify Healthy, Demodicosis, Ringworm, and Inflammatory Infection.

5. CLOUD, MOBILE APP AND WEARABLE GLOVE INTEGRATION: Final results are uploaded to ThingSpeak Cloud, accessed through a MIT App Inventor mobile app, and the entire system is finally integrated into a wearable smart glove for real-time monitoring.

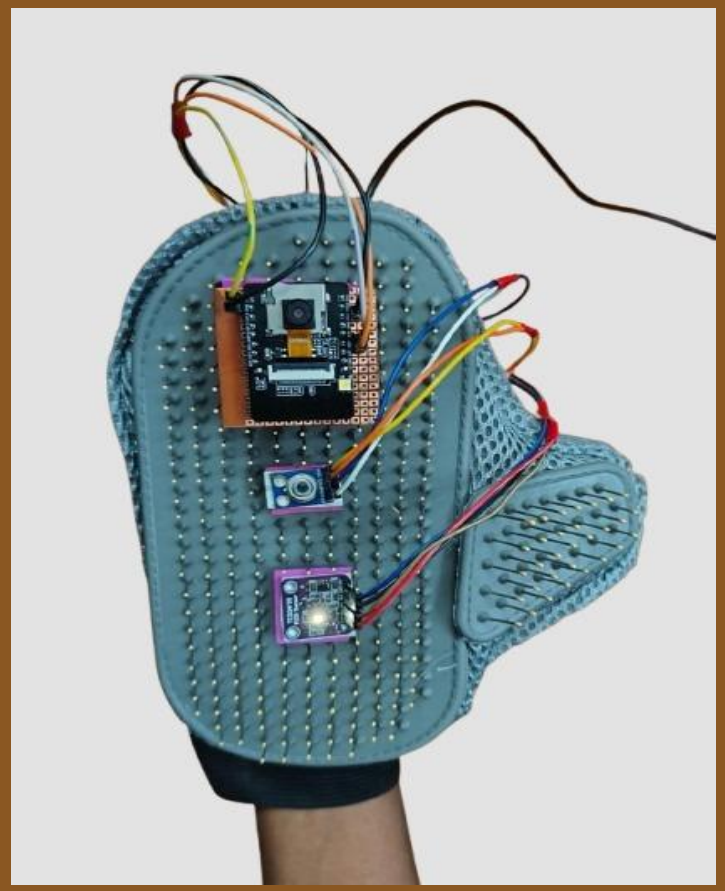
HARDWARE IMPLEMENTATION

- **ESP32-CAM Module Implementation :** We have implemented ESP32-CAM for capturing real-time dog images for wound and skin condition analysis.
- **ESP32 DevKit V1 Integration :** We have used ESP32 DevKit V1 as the main controller for handling and processing sensor data
- **Sensor Integration:** We have integrated temperature, humidity, and color sensors to collect real-time physiological and environmental data.
- **RGB LED Indication System :** We have implemented threshold-based RGB LED indication to show normal and abnormal conditions.
- **Wearable Smart Glove Development:** We have designed and implemented a wearable smart glove by integrating all hardware components for real-time monitoring.

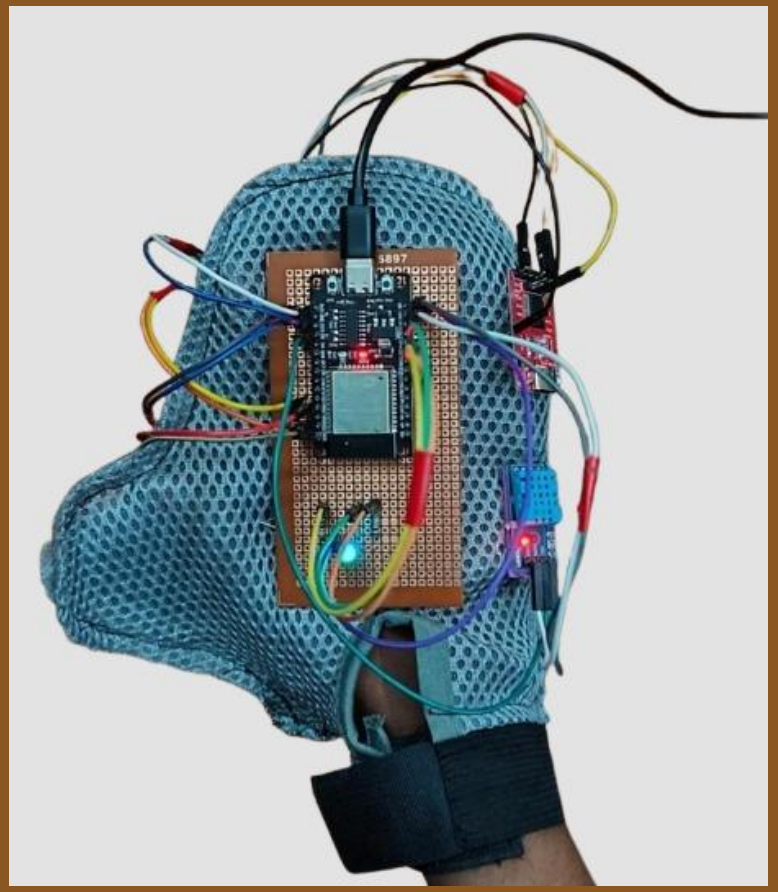
RESULT

SMART GLOVE PROTOTYPE

FRONT VIEW



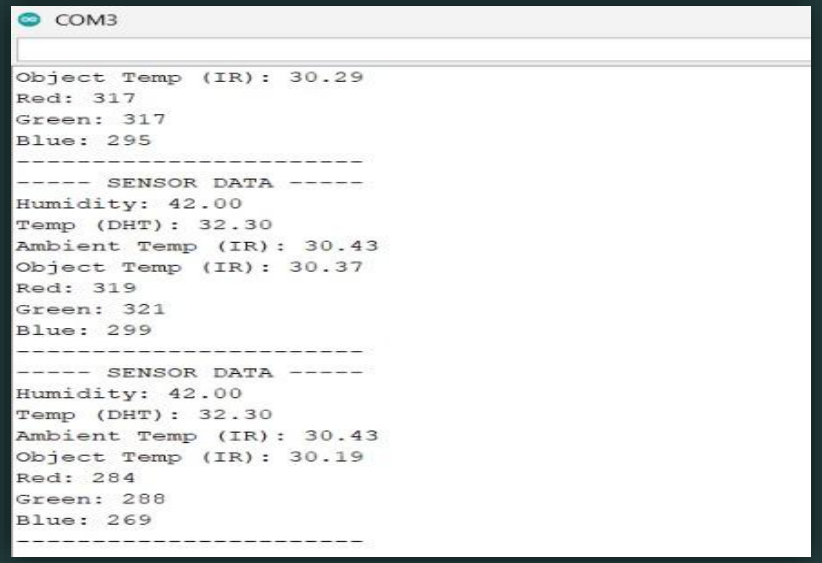
BACK VIEW



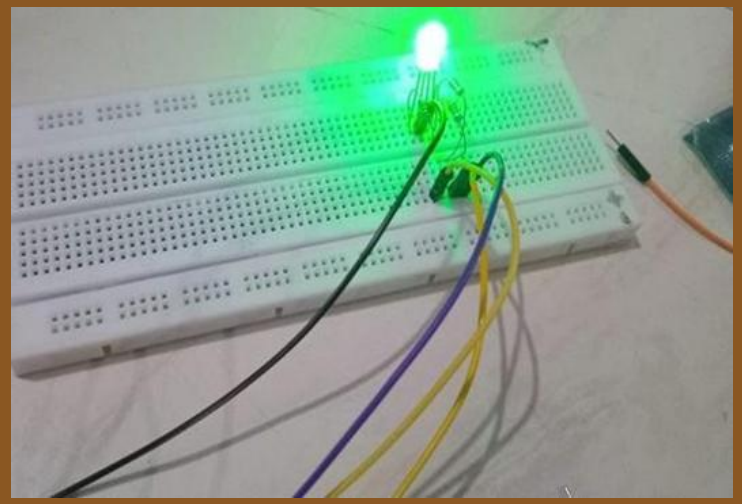
THRESHOLD AND SENSOR READINGS

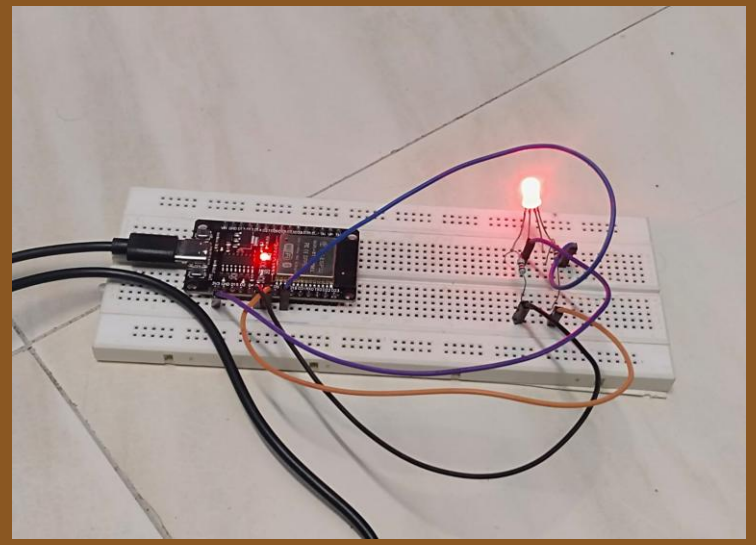
- System accurately detected temperature, humidity, and color changes indicating possible infection
- Instant LED alerts were triggered for abnormal conditions with real-time monitoring
- ESP32-CAM successfully captured and streamed wound images via IP address

PARAMETER	NORMAL RANGE	ABNORMAL CONDITION	ALERT TYPE
Temperature	30 to 36 degree	>38 degree	RED LED
Humidity	40% – 60%	<30 % or >70%	Blue LED
Color	Normal tone R/G/B	Deviation	Alert Signal

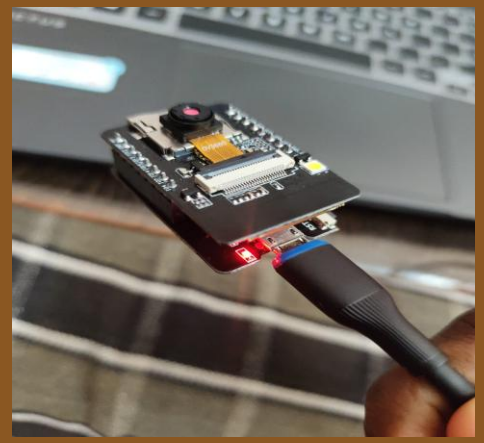


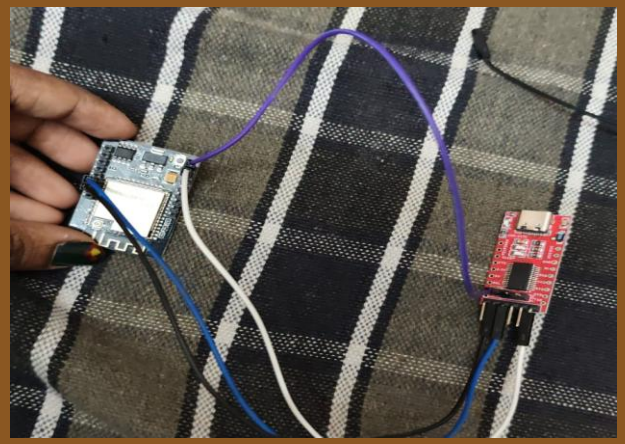
RGB LED

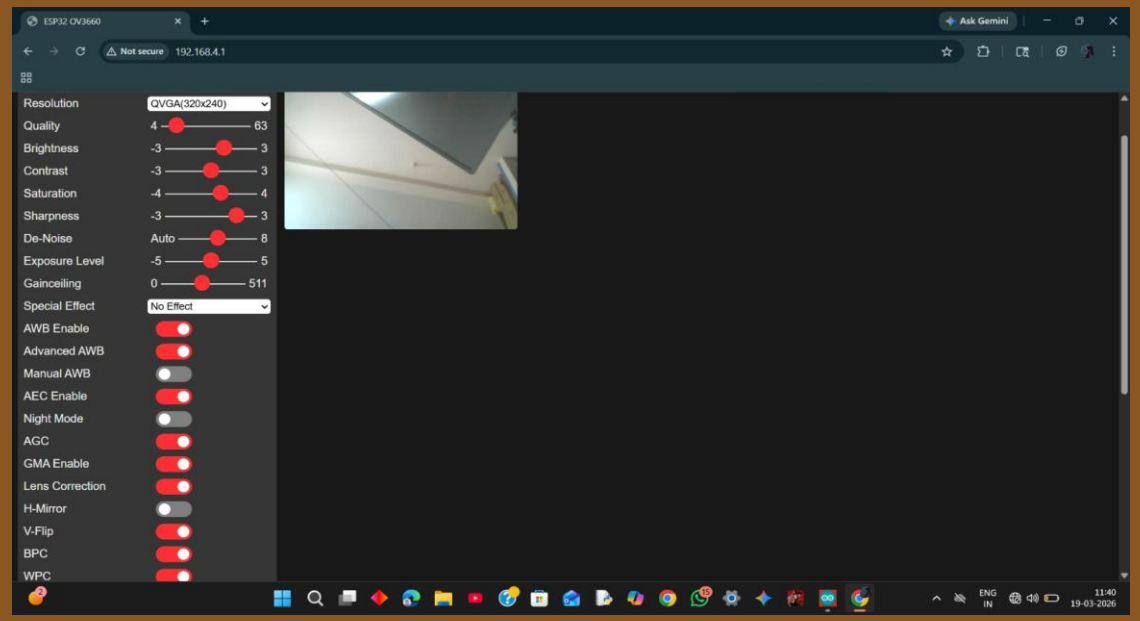




ESP32 CAM





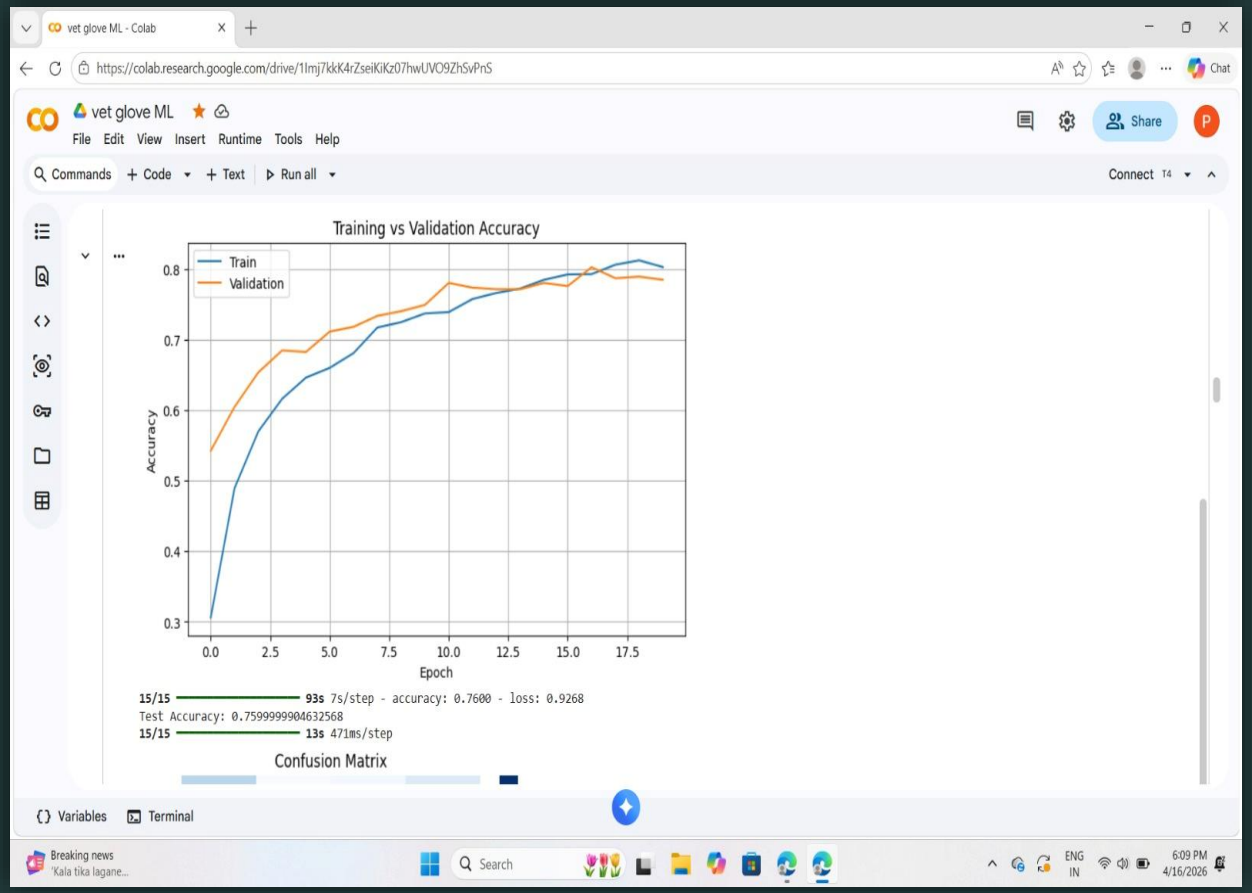


SOFTWARE IMPLEMENTATION

- **ESP32 Programming:** We have programmed ESP32-CAM and ESP32 DevKit V1 using Arduino IDE for image capture and sensor data handling.
- **Image Processing:** We have processed captured images using Python in VS Code with noise removal, resizing, and thresholding techniques.
- **Machine Learning Model :** We have trained and tested a model using mobilenet in Google Colab to classify Healthy, Demodicosis, Ringworm, and Inflammatory Infection.
- **Data Handling:** We have stored images in Google Drive and sensor readings in CSV format for preprocessing and integration.
- **Cloud & Mobile App Integration:** We have used ThingSpeak Cloud for data upload and MIT App Inventor for real-time mobile app visualization.

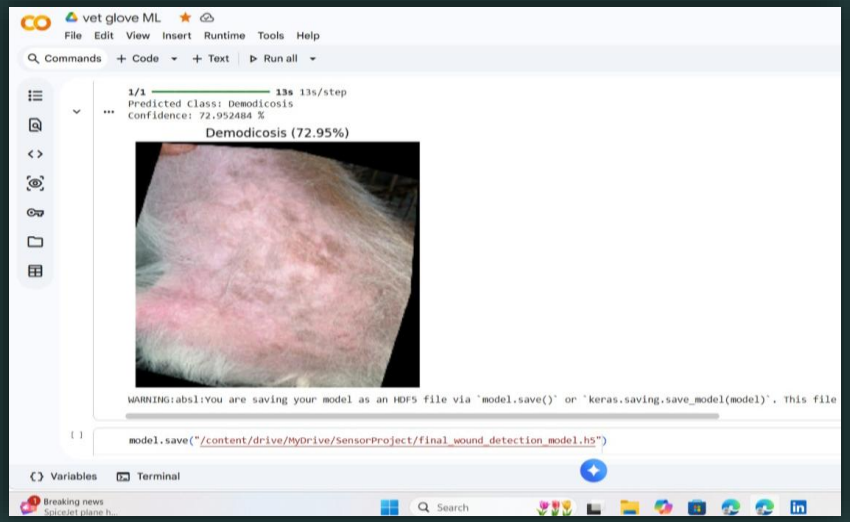
RESULT

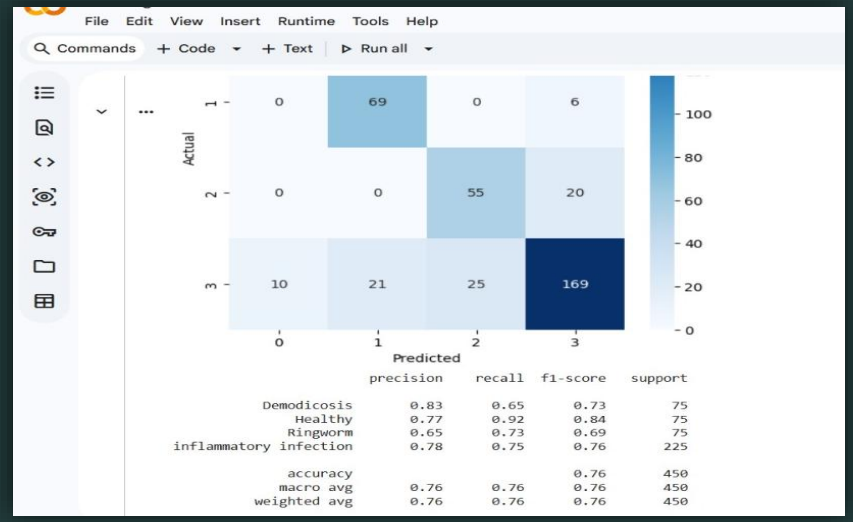
TRAINING AND VALIDATION ACCURACY



EVALUATION METRIC	VALUE
TRAINING ACCURACY	79.25%
VALIDATION ACCURACY	80.22%
TEST ACCURACY	76%
TRAINING LOSS	0.8050
VALIDATION LOSS	0.9113
LEARNING RATE	1.0000e-04
WEIGHTED AVERAGE F1-SCORE	0.76

TEST PREDICTION AND CONFUSION MATRIX



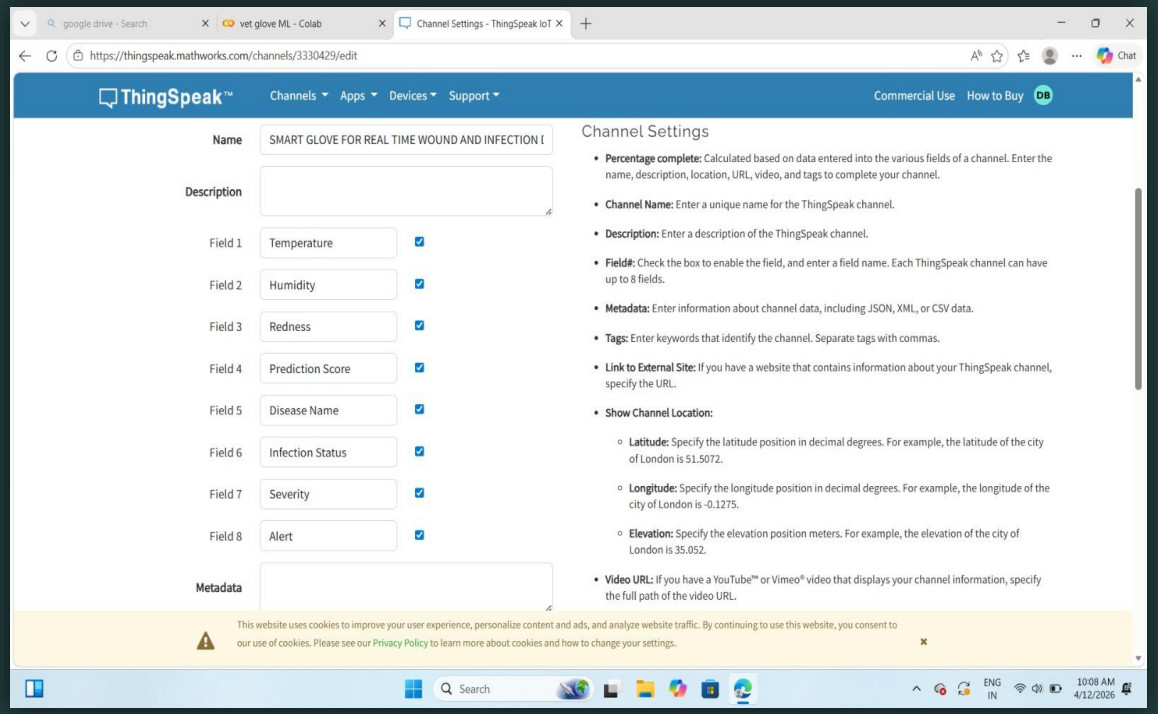


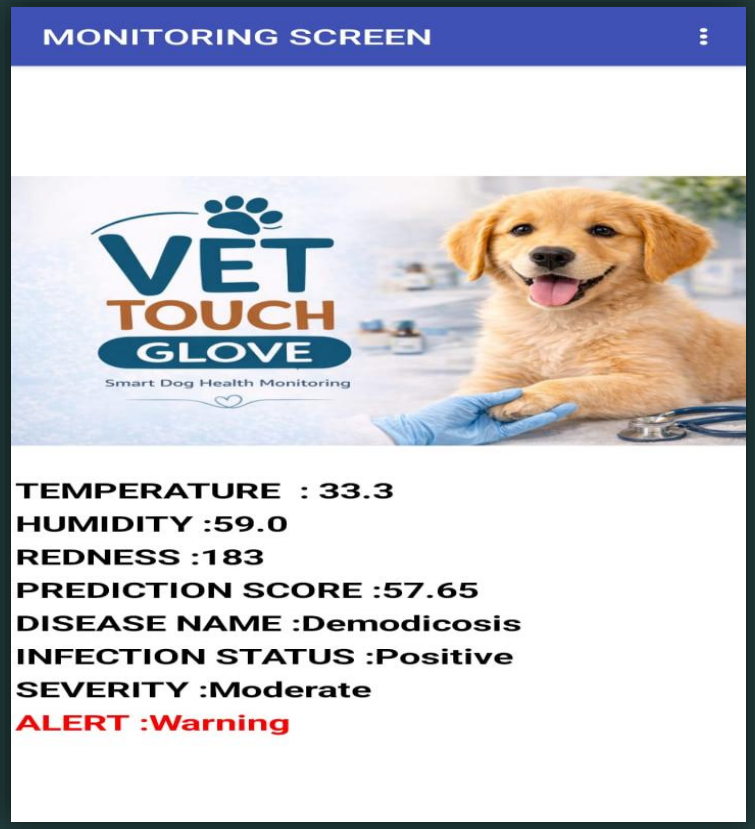
We have tested the system using captured image, sensor readings, and pretrained MobileNet model, and it successfully predicted the condition as Demodicosis.

The model exhibits a classification overlap between Ringworm and Inflammatory infection due to shared feature characteristics, while maintaining its highest precision in distinguishing Healthy subjects from diseased ones.

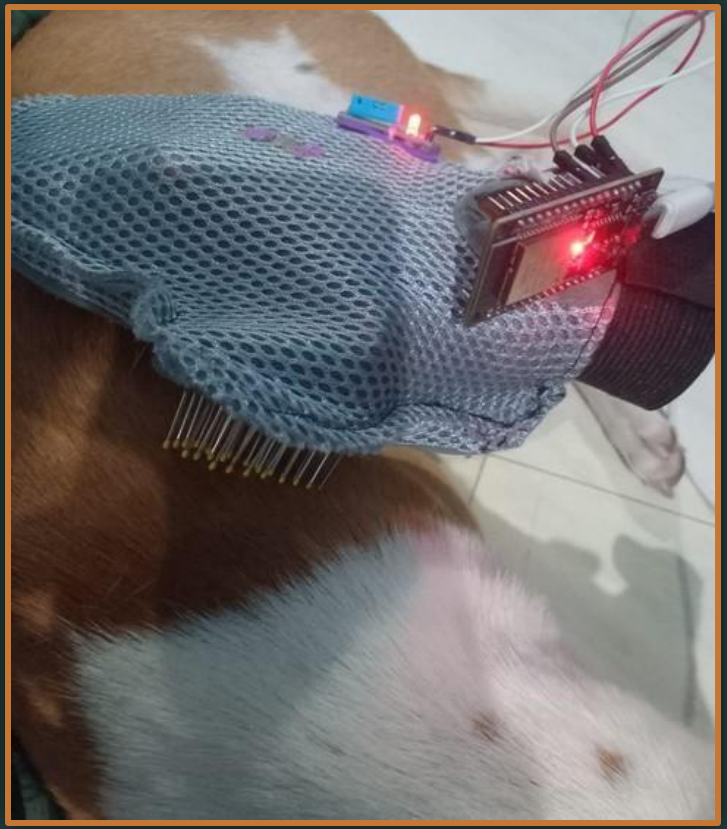
CLOUD AND MOBILE APP DISPLAY OUTPUT

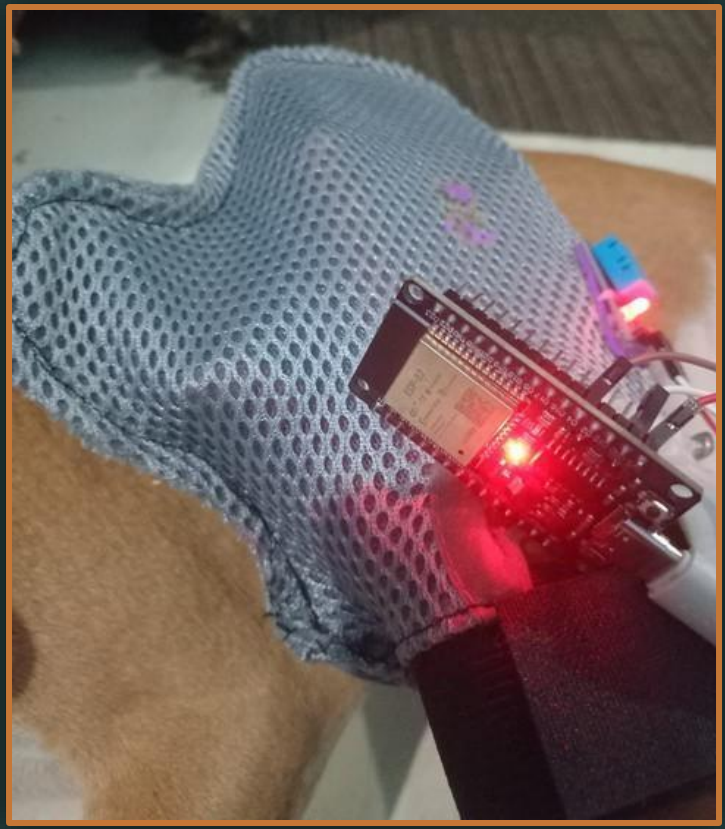
We have integrated ThingSpeak Cloud with the mobile application to fetch real-time sensor data and prediction results. The app displays the updated information continuously for remote monitoring and easy access.

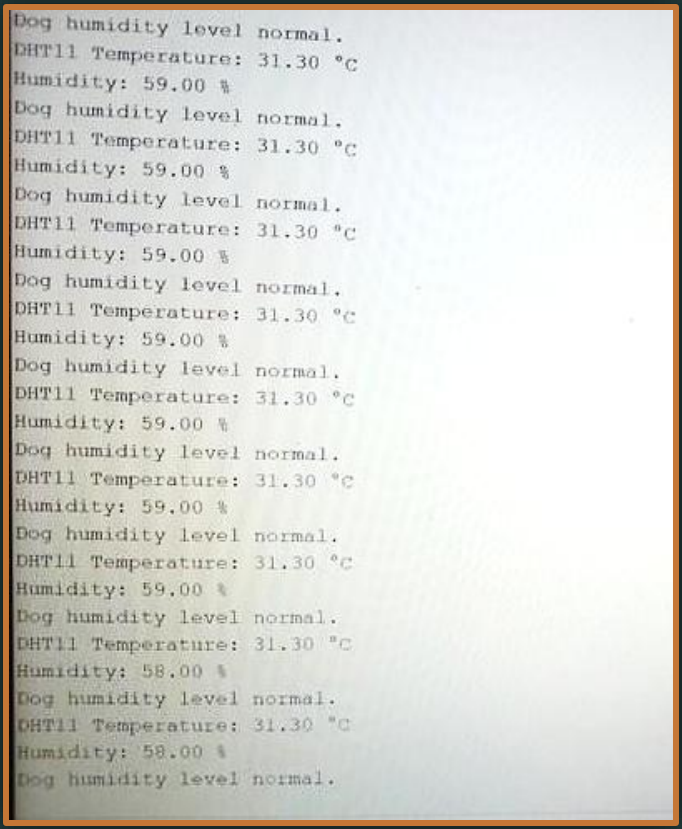




REAL-TIME TESTING AND RESULTS







```
Dog humidity level normal.
DHT11 Temperature: 31.30 °C
Humidity: 59.00 %
Dog humidity level normal.
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Humidity: 59.00 %
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Humidity: 59.00 %
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DHT11 Temperature: 31.30 °C
Humidity: 58.00 %
Dog humidity level normal.
DHT11 Temperature: 31.30 °C
Humidity: 58.00 %
Dog humidity level normal.
```

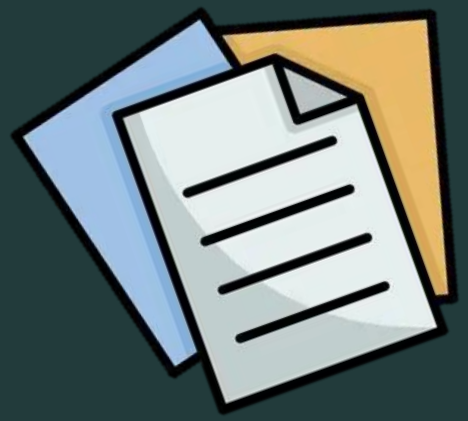
FUTURE SCOPE

- The system can be improved by increasing dataset size and using advanced deep learning models for better accuracy.
- Real-time monitoring can be extended for continuous animal health tracking using the wearable glove.
- The model can be expanded to detect more skin and wound disease categories.
- Mobile app can be enhanced with alerts, history tracking, and veterinary support features.
- Edge AI can be implemented for faster on-device processing without full cloud dependency

CONCLUSION

In this project, a smart wearable glove system has been successfully developed for real-time detection and classification of dog skin and wound diseases. The system integrates ESP32-CAM for image capture and ESP32 DevKit V1 for sensor data collection. Machine learning with MobileNet is used to classify diseases accurately based on processed images. The results are displayed through a mobile application and monitored using ThingSpeak cloud for real-time access. Overall, the system provides an efficient, low-cost, and reliable solution for early disease detection and monitorin.

REFERENCES



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THANK YOU